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Clay Mathematics Institute
Scientific Advisory Board

Dear Members of the Board,

We write to present a solution to the Navier–Stokes Millennium Problem, in a form which differs from the one the official statement anticipates, and to explain why we believe the difference is necessary rather than evasive. We enclose the article *Reformulation and Solution of the Clay Navier–Stokes Problem*.

What we claim. Two things, offered together and standing or falling separately.

First, a **reformulation**. We ask for *constructive existence and qualitative uniqueness of weak solutions to the incompressible Navier–Stokes equations at high Reynolds number*, for given data: a domain $\Omega \subset \mathbb{R}^3$, initial data, a forcing, and boundary conditions. By *constructive* existence we mean existence established by exhibition — a solution produced, with the mesh, the discretisation, the measured residual, the computed stability factors and, for each output, a value with a bound on its error. By *qualitative* uniqueness we mean that the data determine the flow structure and the mean-value outputs, with the line between what is determined and what is not drawn quantitatively, functional by functional, rather than asserted.

Second, a **solution of that problem**: computed weak solutions for specific data, with the error in each output bounded by duality against a computed solution of the adjoint equations, and with mean drag and lift converging to a few per cent on meshes on which the velocity field does not converge and never will. The computational record is in the literature and is cited in the article.

Why not the official formulation. The official statement offers four alternatives, and asks whether smooth data give a solution that stays smooth or whether the velocity blows up. The flows of physical interest are neither. They are turbulent: not smooth, since velocity gradients grow like $\nu^{-1/2}$ and are enormous at any Reynolds number of interest, and not singular, since the velocity stays bounded. A dichotomy which omits the only case that occurs cannot be answered in a way that says anything about fluids.

It is worth putting the point at its strongest, since it is not a statement about which solutions are observed. A high Reynolds number flow set going laminar develops shear instabilities, and these steepen the velocity gradients they feed on until the gradients are sharp enough for viscosity to act — order $\nu^{-1/2}$, on structures of order $\nu^{3/4}$. The laminar state therefore has no continuation: it cannot persist, because the instability grows; it cannot escape by blowing up, because the steepening is arrested as soon as viscosity bites; and it cannot dissipate in the

laminar state what the large scales supply to it. One route remains, and the flow takes it. It becomes turbulent in order to continue to exist. Turbulence is thus not a pathology of the equations but the mode in which their solutions exist at high Reynolds number, and it is the case the official formulation treats as failure.

We would add an observation about mathematical practice. A theorem and its proof are constructed in parallel; the statement is adjusted as the argument develops until the two fit. A formulation fixed in advance of any proof, by a third party, and left standing for twenty-five years is therefore an unusual object, and one should not be surprised if what eventually satisfies it is an argument built to fit rather than a discovery about fluids.

The announced solution, and why it does not close the matter. We take the September 2026 solution announced by OpenAI to be a serious piece of work and we do not dispute it. We observe two things about it.

Its force is not a physical forcing mechanism but is constructed from the residual of the equations themselves, reverse-engineered to match a prescribed singular solution while remaining inside the admissible class. And the statement it proves, (C), is not the negation of (A): (A) and (B) are posed with $f \equiv 0$ and quantified over all admissible data, while (C) and (D) ask for the existence of data *together with an admissible forcing* for which smoothness fails. A blowup driven by an external force shows that something can be done *to* a fluid; (A) asks whether a fluid does it to itself. So the prize may be won without the question everyone means by the Navier–Stokes problem being settled.

There is also a quantitative point. Navier–Stokes rescales, so one unit-viscosity blowup generates the whole family, and the exponents follow from which normalisation is imposed. Held to a fixed admissible forcing class — which is the case that bears on the official statement, since it is that class the formulation fixes — the singular solution has amplitude $\nu^{1/3}$ on support $\nu^{2/3}$, so the family tends to $u \equiv 0$ and the singularity vanishes in the limit where fluids become turbulent. Held instead at fixed amplitude, the force diverges like ν^{-1} and leaves the class. A singularity present at every fixed viscosity and absent as $\nu \rightarrow 0$ is a property of the admissible class rather than of the flow.

What we are and are not asking. We are aware that the prize is governed by conditions we do not here satisfy — publication in a refereed journal and two years of acceptance — and we make no claim under them. Nor do we ask the Board to rule that the official problem is ill-posed. We ask something narrower: that the reformulation be considered on its merits as the question the equations actually raise at high Reynolds number, and the solution on its merits as an answer to it. If the Board judges the official formulation sound, the reformulation still stands as a separate problem with a separate answer, and we would welcome that verdict in preference to silence.

We would be glad to supply computational detail, or to discuss the formulation with anyone the Board cares to nominate.

Yours faithfully,

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